RAJALAKSHMI ENGINEERING COLLEGE

Yugesh Kumar S 251701063-2025-CSD Y2

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|---|---|
| **Started on** | Monday, 28 September 2026, 4:14 PM |
| **State** | Finished |
| **Completed on** | Wednesday, 30 September 2026, 6:43 AM |
| **Time taken** | 1 day 14 hours |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
int main()
{
    int a[100], m;
    int low, high, mid, firstZero;
    scanf("%d", &m);

    for(int i = 0; i < m; i++)
        scanf("%d", &a[i]);

    low = 0;
    high = m - 1;
    firstZero = m;

    while(low <= high)
    {
        mid = (low + high) / 2;
        if(a[mid] == 0)
        {
            firstZero = mid;
            high = mid - 1;
        }
        else
        {
            low = mid + 1;
        }
    }

    printf("%d", m - firstZero);
    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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